FACTORY DOWNTIME ANALYSIS DASHBOARD

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Objective & Scope

Objective

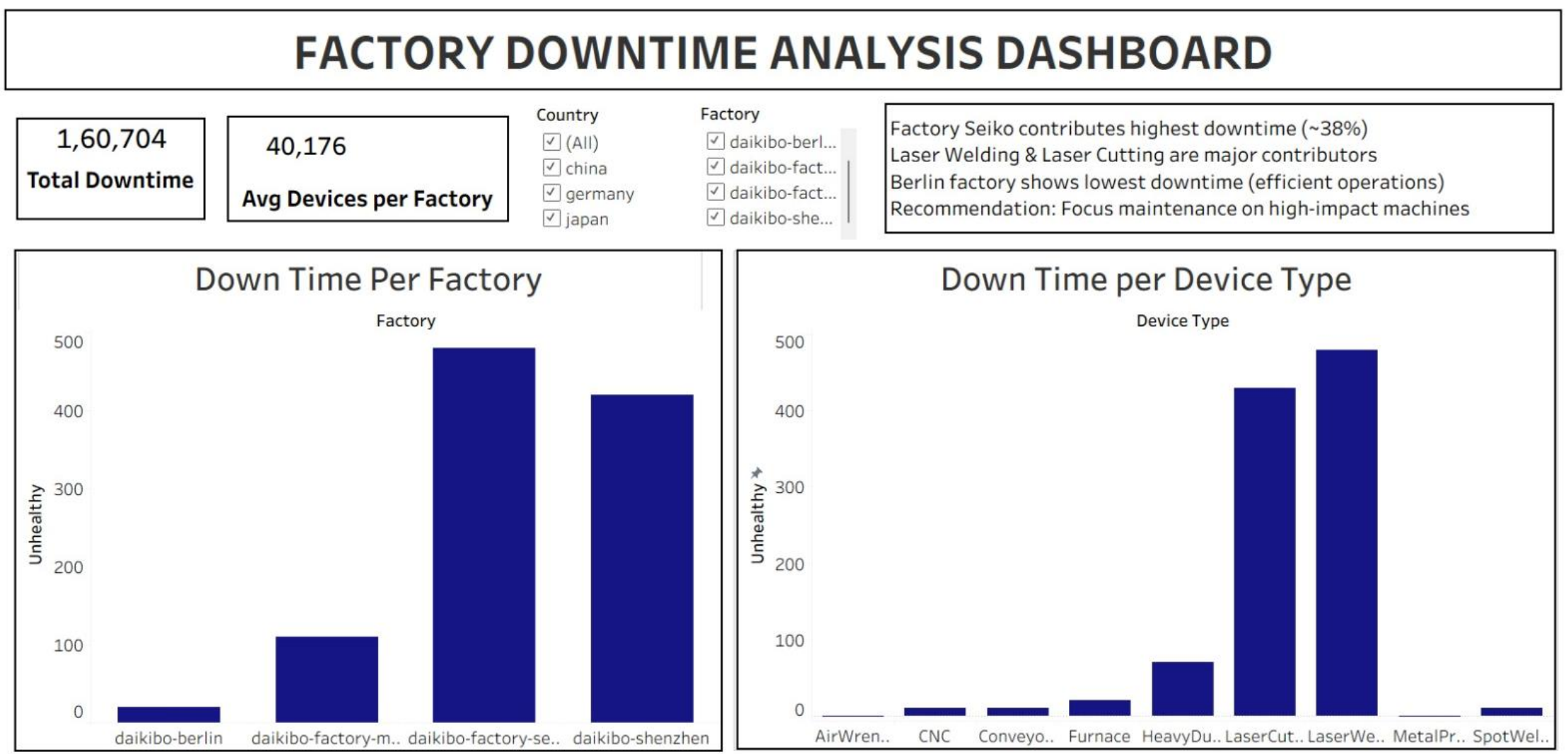
The objective of this project is to analyze factory downtime data using Tableau to identify key factors contributing to operational inefficiencies.

Scope of Analysis

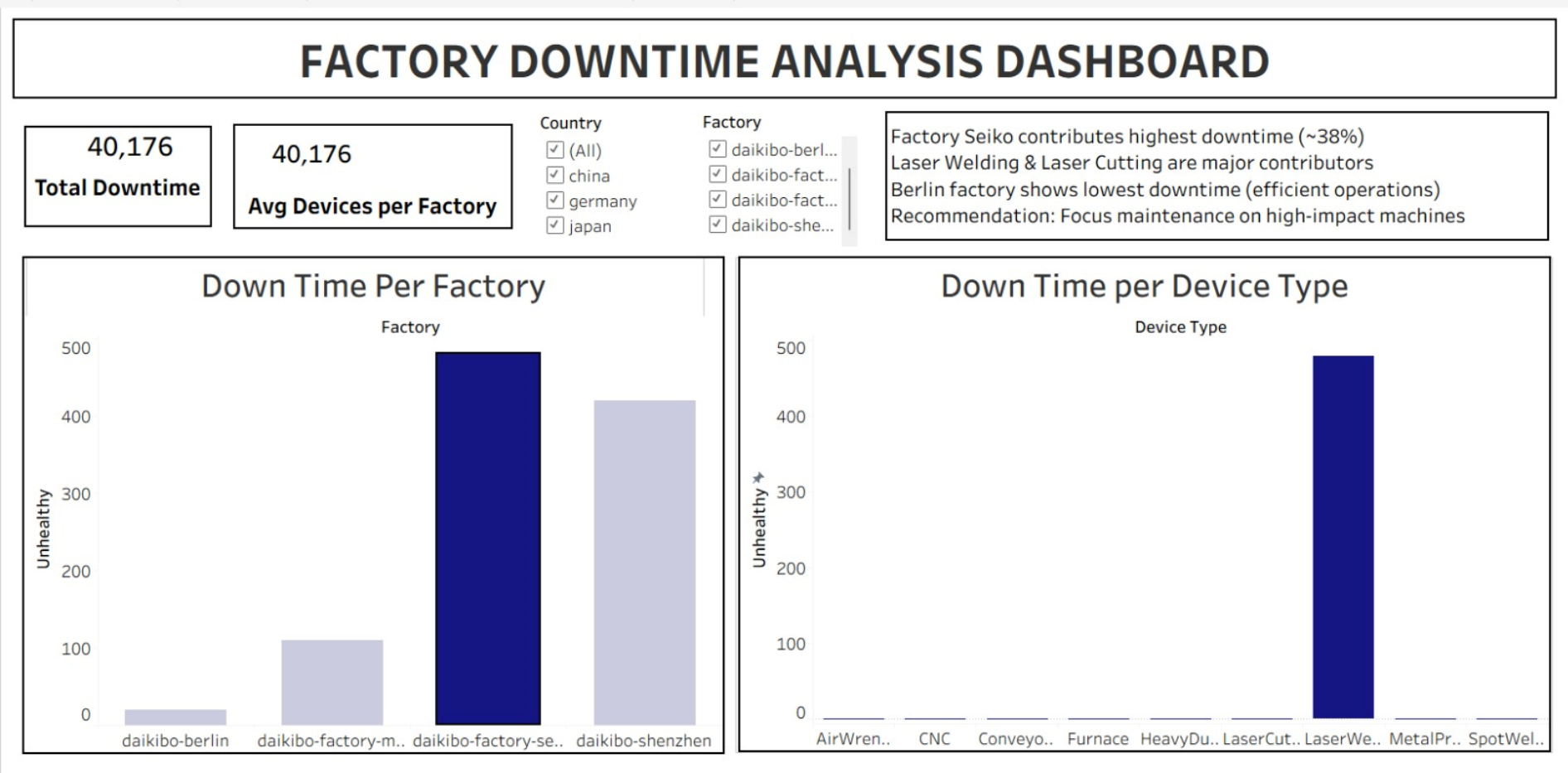
- Analyze downtime across different factory locations
- Identify machine types contributing to maximum downtime
- Develop an interactive dashboard for visualization
- Derive actionable insights to support data-driven decision-making

Dashboard Overview

This dashboard provides a comprehensive analysis of factory downtime across multiple locations and device types. It highlights key performance indicators, identifies major contributors to downtime, and supports data-driven decision-making for operational improvements.



Task: Identify Factory with Highest Downtime



Task Execution

The factory with the highest downtime was identified by analyzing the dashboard.
Factory Seiko was found to have the maximum downtime and was selected to highlight its impact on overall performance.
The dashboard interaction allows dynamic filtering, helping to analyze related device-level downtime.

Questions & Findings

Q1: In which location did machines break the most?

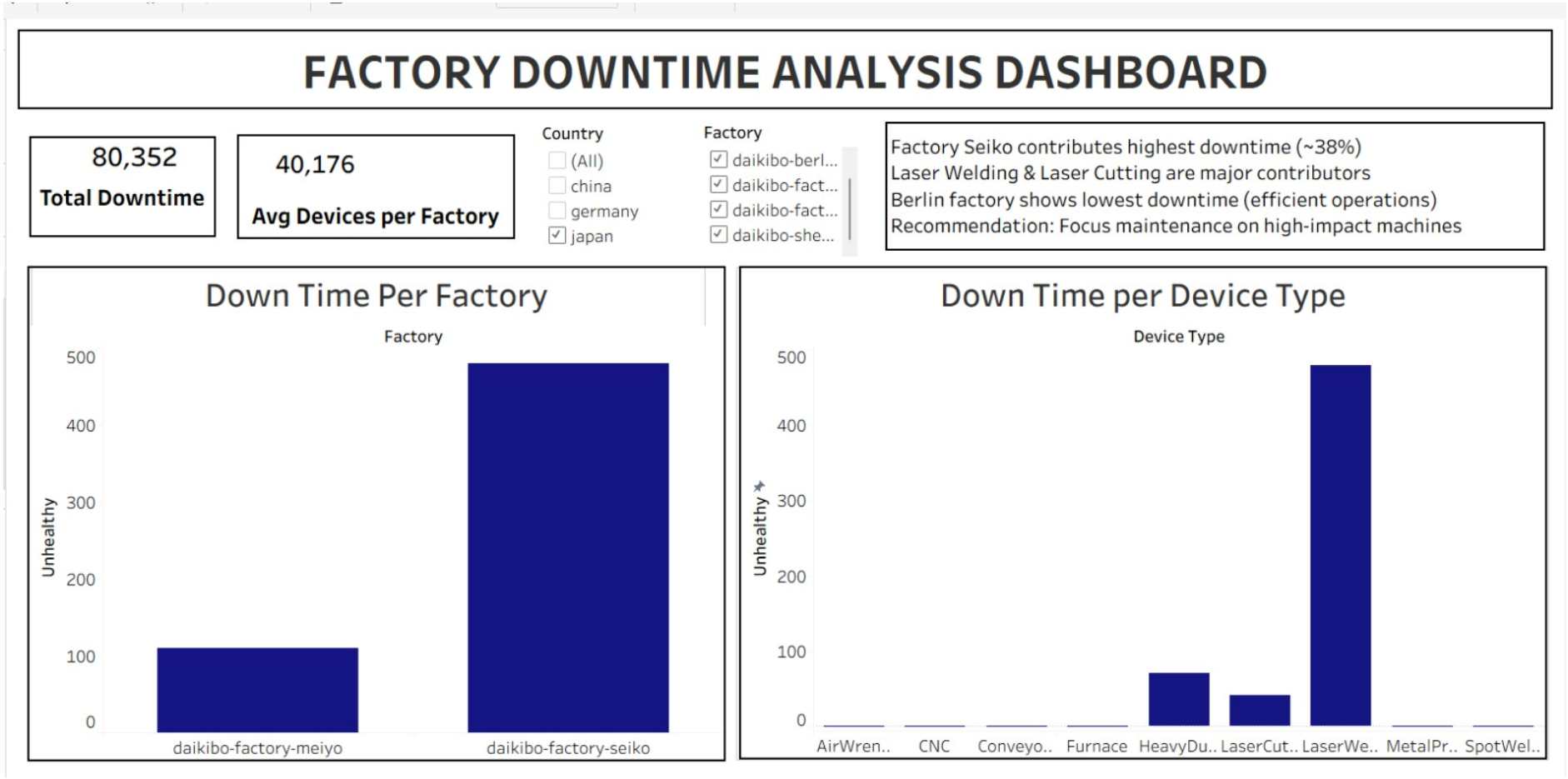
Osaka, Japan recorded the highest machine downtime among all locations.

Q2: What are the machines that broke most often in that location?

Laser Welding and heavy duty drill machines experienced the highest breakdown frequency in Factory Seiko at Osaka, Japan.

Insight:

The high downtime at Japan is primarily driven by specific machine types, indicating a need for targeted maintenance and operational improvements.



Key Insights

- Factory Seiko contributes the highest share of downtime (~38%), indicating a critical operational bottleneck.
- Laser Welding and Laser Cutting are the primary drivers of downtime across factories.
- Shenzhen factory shows moderate downtime levels and requires monitoring.
- Berlin factory demonstrates minimal downtime and strong operational efficiency.
- Recommendation: Focus maintenance on high-impact machines

Conclusion & Strategic Recommendations

- Factory Seiko is identified as the largest contributor to overall downtime, indicating a critical operational bottleneck that requires immediate attention.
- Laser Welding and Laser Cutting devices are the primary drivers of downtime, highlighting the need for targeted maintenance and process optimization.
- Berlin factory demonstrates strong operational performance and can be leveraged as a benchmark to replicate best practices across other factories.
- Shenzhen factory reflects moderate downtime levels and should be closely monitored to prevent performance decline.
- A focused strategy on high-impact factories and key device categories will enable reduction in downtime, improved efficiency, and better resource utilization across operations.